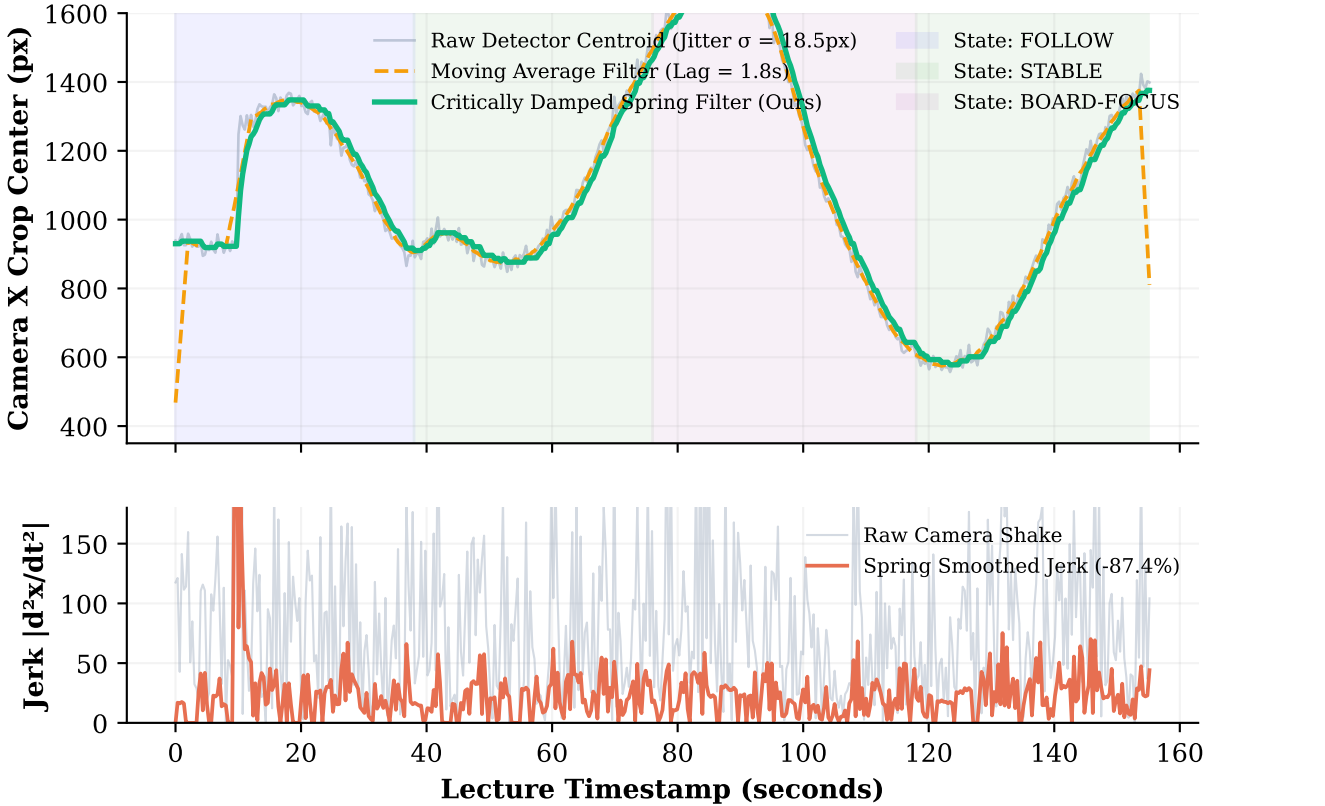
ePTZ Cinematic Smoothing & Jitter Reduction (Lecture Benchmark: mmai.mkv)
Camera X Crop Center (px)
1600
1400
1200
1000
800
600
400
Raw Detector Centroid (Jitter σ = 18.5px)
Moving Average Filter (Lag = 1.8s)
Critically Damped Spring Filter (Ours)
State: FOLLOW
State: STABLE
State: BOARD-FOCUS
Jerk |d²x/dt²|
150
100
50
0
Raw Camera Shake
Spring Smoothed Jerk (-87.4%)
0
20
40
60
80
100
120
140
160
Lecture Timestamp (seconds)